

The trigonometric functions can also be used to solve oblique triangles. Here are the four oblique triangle cases which may be solved.

- law of sines {
law of cosines {
- Case 1: One side and two angles are known (ASA or SAA)
 - Case 2: Two sides and the angle opposite one of those sides (SSA)
 - Case 3: Two sides and the included angle (SAS)
 - Case 4: Three sides (SSS)

I. The Law of Sines

The Law of Sines says that in any triangle the lengths of the sides are proportional to the sines of the corresponding opposite angles.

We use the Law of Sines to solve triangles in Cases 1 and 2.

For a triangle ABC , state the Law of Sines.

Finding an angle

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Finding a side

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

II. The Ambiguous Case

The triangle situation given in Case 2, in which two sides and the angle opposite one of those sides are known, is sometimes called the **ambiguous case** because there may be two triangles, one triangle, or no triangle with the given properties. In general, if $\sin A < 1$, we must check the angle and its supplement as possibilities, because any angle smaller than 180° can be in the triangle. To decide whether either possibility works, check to see whether the resulting sum of the angles exceeds 180° . It can happen that both possibilities are compatible with the given information. In that case, two different triangles are solutions to the problem.

When you are given a SSA triangle one of the following will be true.

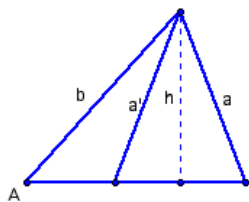
1. No solution – No triangle
2. Exactly one solution – One Triangle
3. Two solutions – Two Triangles

other given side
 $b \sin A$
given angle

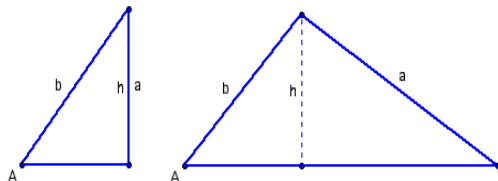
In an SSA triangle we must first find the height (h) of the triangle. The height of the triangle is $h = b \sin A$. The possible solutions depend on whether the given angle is acute or obtuse.

Case I: $A < 90^\circ \rightarrow$ Acute Angle

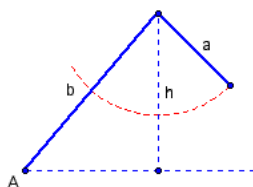
1. If $b > a > b \sin A$, there are two solutions, two triangles
 $b > a > h$



2. If $a > b > b \sin A$, there is one solution, one triangle
 $a > b > h$

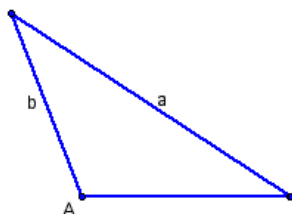


3. If $a < b \sin A < b$, there is no solution, no triangle
 $a < h < b$

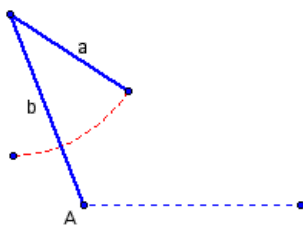


Case II: $A \geq 90^\circ$ --Obtuse Angle

1. If $a > b$, there is one solution, one triangle



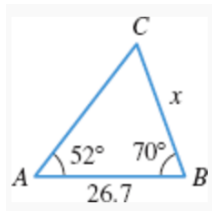
2. If $a \leq b$, there is no solution, no triangle



Example 1: Use the Law of Sines to find the indicated side x or angle θ .

$$\angle C = 180 - (52^\circ + 70^\circ) = 58^\circ$$

a.

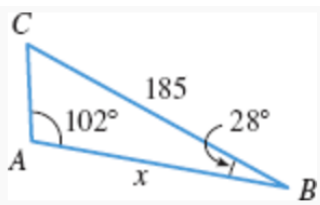


$$\frac{x}{\sin 52^\circ} = \frac{26.7}{\sin 58^\circ}$$

$$x = \frac{26.7 \sin 52^\circ}{\sin 58^\circ}$$

$$= 24.8$$

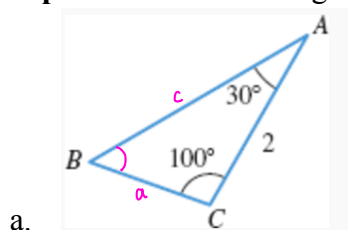
b.



$$\frac{x}{\sin 50} = \frac{185}{\sin 102}$$

$$x \approx 144.9$$

Example 2: Solve the triangle using the Law of Sines.

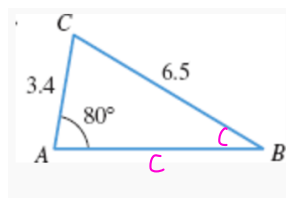


$$\angle B = 180^\circ - 130^\circ = \boxed{50^\circ}$$

$$\frac{a}{\sin 30^\circ} = \frac{2}{\sin 50^\circ}$$

$$a = \frac{(2 \sin 30^\circ)}{\sin 50^\circ} = \boxed{1.3}$$

$$c = \frac{(2 \sin 100^\circ)}{\sin 50^\circ} = \boxed{2.6}$$



$$\frac{\sin 80^\circ}{6.5} = \frac{\sin B}{3.4}$$

$$B = \sin^{-1} \frac{3.4 \sin 80^\circ}{6.5}$$

$$\angle B \approx \boxed{31.0^\circ}$$

$$\angle C = 180^\circ - 80^\circ - \angle B$$

$$\angle C \approx \boxed{69.0^\circ}$$

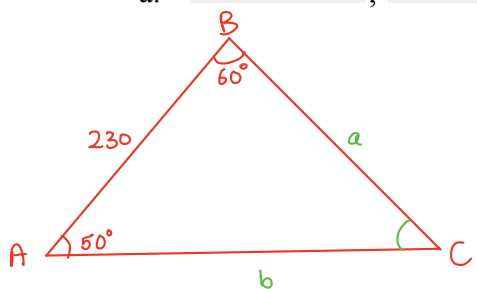
$$\frac{6.5}{\sin 80^\circ} = \frac{c}{\sin \angle C}$$

$$c = \frac{6.5 \sin \angle C}{\sin 80^\circ}$$

$$\boxed{c \approx 6.2}$$

Example 3: Sketch the triangle, and then solve it using the Law of Sines.

a. $\angle A = 50^\circ$, $\angle B = 68^\circ$, $c = 230$



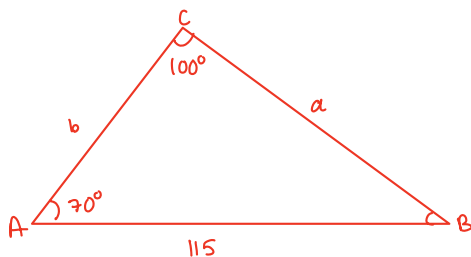
$$\angle C = \boxed{62^\circ}$$

$$\frac{b}{\sin 68^\circ} = \frac{230}{\sin 62^\circ}$$

$$b = \frac{(230 \sin 68^\circ)}{\sin 62^\circ} = \boxed{241.5}$$

$$a = \frac{(230 \sin 50^\circ)}{\sin 62^\circ} = \boxed{199.5}$$

b. $\angle B = 10^\circ$, $\angle C = 100^\circ$, $c = 115$



$$\angle B = 180^\circ - 100^\circ - 70^\circ = \boxed{10^\circ}$$

$$\frac{a}{\sin 70^\circ} = \frac{115}{\sin 100^\circ}$$

$$a = \frac{(115 \sin 70^\circ)}{\sin 100^\circ}$$

$$\boxed{a \approx 109.7}$$

$$\frac{b}{\sin 10^\circ} = \frac{115}{\sin 100^\circ}$$

$$b = \frac{115 \sin 10^\circ}{\sin 100^\circ}$$

$$\boxed{b \approx 20.3}$$

Example 4: Use the Law of Sines to solve for all possible triangles that satisfy the given conditions. There may be two triangles, one triangle, or no triangles that satisfy the given conditions.

i. $a = 28, b = 15, \angle A = 110^\circ$

$a > b \rightarrow$ one solution

$$\frac{\sin B}{15} = \frac{\sin 110^\circ}{28}$$

$$\sin B = \frac{15 \sin 110^\circ}{28}$$

$$\angle B = \sin^{-1}\left[\frac{(15 \sin 110^\circ)}{28}\right] = \boxed{30.2^\circ} \rightarrow \text{sto } x$$

$$\angle C = 180^\circ - (x + 110^\circ) = \boxed{39.8^\circ} \rightarrow \text{sto as } x$$

$$\begin{aligned} \frac{c}{\sin x} &= \frac{28}{\sin 110^\circ} \\ c &= \frac{(28 \sin x)}{\sin 110^\circ} \\ &= \boxed{19.1} \end{aligned}$$

ii. $a = 20, c = 45, \angle A = 125^\circ$

$a < b \rightarrow$ no solution

no solution

iii. $b = 25, c = 30, \angle B = 25^\circ$

$$h = 30 \sin 25^\circ$$

$$= 12.68$$

$b > a > h \rightarrow$ 2 solutions

find $\begin{cases} \angle C_1, \angle C_2 \\ \angle A_1, \angle A_2 \\ a_1, a_2 \end{cases}$

$$\angle C_1: \frac{\sin C_1}{30} = \frac{\sin 25^\circ}{25}$$

$$C_1 = \sin^{-1}\left[\frac{(30 \sin 25^\circ)}{25}\right]$$

$$= \boxed{30.5^\circ} \rightarrow \text{sto } x$$

$$\angle C_2 = 180^\circ - x = \boxed{149.5^\circ} \rightarrow \text{sto } x$$

$$\angle A_2 = 180^\circ - (x + 25^\circ) = \boxed{5.5^\circ}$$

$$\angle A_1 = 180^\circ - (25^\circ + \angle C_1) = \boxed{124.5^\circ}$$

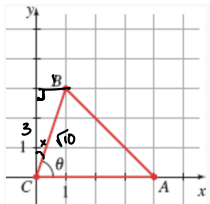
$$\begin{aligned} \frac{a_1}{\sin A_1} &= \frac{25}{\sin 25^\circ} \\ a_1 &= \frac{(25 \sin A_1)}{\sin 25^\circ} = \boxed{48.8} \end{aligned}$$

$$\frac{a_2}{\sin A_2} = \frac{25}{\sin 25^\circ}$$

$$a_2 = \frac{(25 \sin A_2)}{\sin 25^\circ}$$

$$a_2 = \boxed{5.6}$$

Example 5: Use the Law of Sines to find the angle in degrees, rounded to one decimal place.



$$\frac{\sin 90^\circ}{\sqrt{10}} = \frac{\sin x}{1}$$

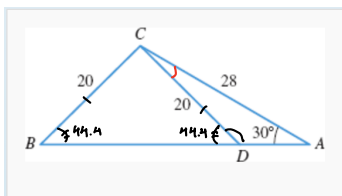
$$x = \sin^{-1}\left[\frac{(\sin 90^\circ \cdot 1)}{\sqrt{10}}\right]$$

$$x \approx 18.4349$$

$$90^\circ - x = \theta$$

$$\theta \approx 71.6^\circ$$

Example 6: Find $\angle DCA$



$$\frac{\sin 30}{20} = \frac{\sin \angle B}{28}$$

$$\angle BDC = \angle B \approx 44.4$$

$$\angle DCA = 180^\circ - 30^\circ - (180^\circ - 44.4^\circ)$$

$$\angle DCA \approx 14.43^\circ$$

Example 7: The path of a satellite orbiting the earth passes directly over two tracking stations A and B , 50 mi apart. When the satellite is on one side of the two stations, the angles of elevation at A and B are 87° and 84.2° , respectively, as shown in the figure.

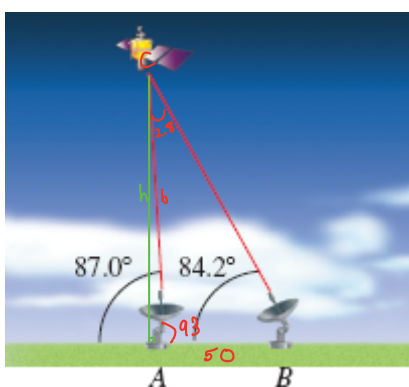
(a) How far is the satellite from station A ?

$$\frac{b}{\sin(84.2^\circ)} = \frac{50}{\sin(2.8^\circ)}$$

$$b = \frac{(50 \sin 84.2^\circ)}{\sin 2.8^\circ}$$

$$b \approx 1018.3 \text{ mi}$$

(b) How high is the satellite above the ground?

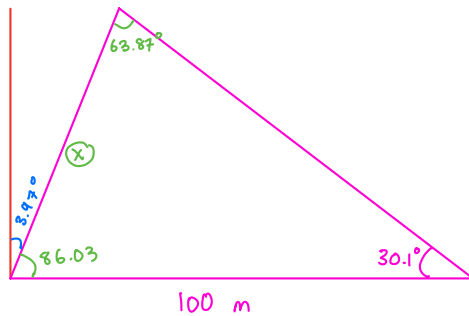


$$\sin 87^\circ = \frac{h}{b}$$

$$h = b \cdot \sin 87^\circ$$

$$h \approx 1016.9 \text{ mi}$$

Example 8: The bell tower of the cathedral in Pisa, Italy, leans 3.97° from the vertical. Two tourists stand 100 m from its base, with the tower leaning directly toward them. It is observed that the angle of elevation to the top of the tower is 30.1° . Find the length of the tower to the nearest meter.



$$\frac{x}{\sin 30.1^\circ} = \frac{100}{\sin 63.87^\circ}$$

$$x = \frac{(100 \sin 30.1^\circ)}{\sin 63.87^\circ} = \boxed{56 \text{ m}}$$